RAG System Report: Parameter Experimentation on Lloyds Banking Group plc Risk Factors Document

Author: Divine Abiri
Date: 29 July 2026
Work Experience: Lloyds Banking Group - AI Work Experience Placement

Overview

Lloyds Bank Risk Factors PDF document is a Form 6-K filing submitted by Lloyds Bank plc to the U.S Securities and Exchange Commission on 29 January 2026. It contains a detailed summary of the potential risk factors that could materially affect Lloyds Group's business operations, financial condition, and prospects; it also covers economic and financial risks, regulatory and legal risks, and business and operational risks(including cybersecurity, AI and ESG matters). The document is 22 pages of dense, unstructured legal and financial prose.

Purpose of the document for the experiment

In most RAG tutorials/demonstrations, the documents typically used are FAQ-style or short-paragraph documents. But this document is a regulatory risk filing containing long, dense paragraphs, with extensive cross-referencing between sections, and specific embedded numeric facts(e.g dates, percentages), which makes it harder. Still, it is a more realistic test case for a RAG system than a toy dataset, and it is directly relevant to the kind of document a Lloyds risk or compliance analyst might need to query.

System Overview

- **Loader**: LangChain PyPDFLoader
- **Chunking (baseline):** LangChain RecursiveCharacterTextSplitter, chunk_size = 1000, chunk_overlap = 20
- **Embeddings:** OpenAI text-embedding-3-small
- **Vector Store:** ChromaDB (persistent)
- **LLM(baseline):** gpt-5.4-mini
- **Full code:** https://github.com/divineabiri/PDF-RAG-System.git

Methodology

- I have created a fixed set of questions that would be used consistently across every experiment so the results are comparable

#	Question	Purpsose
Q1	What was the total provision Lloyds recognised for the motor finance commission redress scheme?	This tests the RAG system's ability to get precise numerical facts.
Q2	What does the document say about COVID-19?	This tests whether the RAG system can locate facts buried within a longer thematic paragraph.
Q3	What are the main risks Lloyds faces from artificial intelligence?	This tests the RAG system's ability to summarise
Q4	What powers does the Bank of England have under the Banking Act 2009?	This tests whether the RAG System can recognise specific legal/regulatory detail
Q5	What is Lloyds' dividend policy for 2026?	This is a planned trap question for the RAG system to test for hallucinations, as it should not give false information that does not exist in the document. The proper response should be "I don't know".

- After completing each test, the results would be evaluated manually through a simple small-scale scoring system.
  Where;
  Correct = right chunk rettrieved, answer accurate.
  Partial = relevant chunk received but answer incomplete/vague.
  Incorrect/Hallucinated = wrong chunk retrieved, or answer made up.

Experiments

- ### Chunking

Experiment 1 - Chunk Size Parameter
- **Changed:** 500/1000/2000 (chunk_overlap fixed at 20)
- **Hypothesis:** I think that the larger the chunk size, the more detailed the result brought back by the LLM.

Results Table

Chunk_size	Q1	Q2	Q3	Q4	Q5
500 = **253 Chunks Created**	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says that the COVID-19 pandemic demonstrated the vulnerability of a highly interlinked global economy.** **Correct**	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI-related risks: amplified existing risks, new operational and competitive risks, falling behind peers if it adopts AI more slowly, and increased cyberattack, data leak, and information disclosure risks. It also needs to protect privacy and ensure ethical data processing to maintain trust and comply with regulations.** **Partial**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers including mandatory transfer powers, the ability to amend contract terms, and to override events of default or termination rights. It can also override, vary, or impose contractual obligations between a UK bank, its holding company, and group undertakings to help a transferee or successor bank operate effectively.** **Correct**	What is Lloyds' dividend policy for 2026? **-> I don't know.** **Correct**
1000 = **126 Chunks**	What was the total provision Lloyds recognised	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic**	What are the main risks Lloyds faces from artificial intelligence?	What powers does the Bank of England have under the Banking Act 2009?	What is Lloyds' dividend policy for 2026? **-> I don't know.**

Created	for the motor finance commission redress scheme? **-> £1,950 million** Correct	**triggered disruption to global supply chains at a scale not previously experienced.** Correct	**-> Lloyds faces several main AI-related risks: amplified existing risks and new operational and competitive risks; privacy and ethical data-processing risks; falling behind peers and losing business opportunities; legal and regulatory uncertainty across jurisdictions; increased cyberattack, data leak, and information-loss risks; and risks of biased or harmful outcomes such as discrimination or misidentifying vulnerable customers.** Correct	**-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing or likely failing UK-incorporated deposit-takers. These include stabilisation options such as transfer to a private purchaser, transfer to a bridge bank, transfer to an asset management vehicle, bail-in, and temporary public ownership. It also has powers to amend contract terms, override default or termination rights, and override, vary, or impose contractual obligations to help a transferee or successor bank operate effectively.** Correct	Correct
2000 = **68 Chunks Created**	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** Correct	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: operational and competitive risk from falling behind peers, legal and regulatory risk from rapidly changing AI rules, cybersecurity and data leakage risks, model risks like bias and errors, IP copyright and malicious-use risks, colleague misuse risk, customer harm risks such as discrimination or misidentifying vulnerable customers, and added governance challenges from agentic AI.** Correct	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers including:** **- a mandatory write-down and conversion power for Tier 1 and Tier 2 capital instruments** **- a bail-in power for capital instruments and most unsecured liabilities** **- powers to modify contractual arrangements, override certain default or termination rights, and vary contractual obligations within a banking group** Correct	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct

Analysis: A chunk size of **1000** gave the best balance between context and precision. Chunks of **500** sometimes split related legal information, while **2000** often included unnecessary context because the document contains long regulatory paragraphs.

Experiment 2 - Chunk Overlap

- **Changed:** 0/20/100
- **Hypothesis:** The context of the question asked would be gathered more as the chunk overlap increases

Result Table

Chunk_overlap	Q1	Q2	Q3	Q4	Q5
0	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** Correct	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI-related risks: amplified existing risks, new operational and competitive risks, privacy and ethical data-processing risks, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle**	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct

			misidentification, governance challenges from agentic AI, and increased legal/regulatory and cybersecurity risks. **Partial**	**- bailing in equity, capital instruments, and liabilities** **- modifying or overriding contractual arrangements, including contract terms and default/termination rights** **I don't know the rest of the powers from the provided context.** **Correct**	
20	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** **Correct**	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection issues, ethical processing concerns, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer misidentification, governance challenges from agentic AI, increased cyberattack and data leak risk, and evolving legal/regulatory risk from changing AI rules.** **Correct**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including mandatory transfer of business or shares, transfer to a bridge bank, transfer to an asset management vehicle, bail-in of equity/capital instruments/liabilities, and temporary public ownership. It can also modify contractual arrangements, amend contract terms, override default or termination rights, and impose or vary obligations between a bank and its group.** **Correct**	What is Lloyds' dividend policy for 2026? **-> I don't know.** **Correct**
100	What was the total provision Lloyds recognised for the motor finance commission redress scheme**?** **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says COVID-19 triggered disruption to global supply chains at a scale not previously experienced.** **Correct**	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing risks and new operational and competitive risks: privacy, ethical processing, and data protection compliance risks; falling behind peers and losing business opportunities; discrimination and vulnerable customer misidentification; governance challenges, especially with agentic AI; increased legal and regulatory risk from rapidly changing AI rules; and higher cyberattack, data leak, financial, and reputational risks.** **Correct**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private purchaser** **- transferring business to a bridge bank** **- transferring business to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- temporarily taking a bank into public ownership** **It also has powers to modify or override contractual arrangements, including varying debt terms and overriding default or termination rights.** **Correct**	What is Lloyds' dividend policy for 2026? **-> I don't know.** **Correct**

Analysis: Changing the overlap had little impact on accuracy. An overlap of 20 preserved enough context between chunks, while increasing it to 100 mainly increased duplicated information without improving results.

Experiment 3 - Splitter Type

- **Changed:** manual fixed character **split_text_into_chunks** vs **RecursiveCharacterTextSplitter**
- **Hypothesis:** They might function in the same manner

Result Table

Splitter	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**
split_text_into_chunks	What was the total provision	What does the document say about COVID-19?	What are the main risks Lloyds faces from	What powers does the Bank of	What is Lloyds' dividend policy for 2026?

= **Built 112 chunks (chunk_size= 1000, overlap = 20)**	Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	**-> The document says COVID-19 triggered disruption to global supply chains at a scale not previously experienced, and it also led to remote working arrangements that increase reliance on IT systems and may raise fraud, conduct, operational, and other risks.** **Correct**	artificial intelligence? **-> Lloyds faces several main AI risks: falling behind industry innovation, regulatory and legal uncertainty, cyberattacks and data leaks, bias and model errors, IP/copyright and malicious misuse risks, colleague misuse, customer discrimination or vulnerable customer misidentification, and added governance challenges from agentic AI systems.** **Correct**	England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to override, vary or impose contractual obligations between a UK bank, its holding company and group undertakings, and to amend the terms of contracts, including varying the maturity of debt instruments.** **Correct**	**-> I don't know.** **Correct**
RecursiveCharcterText Splitter= **Built 126 chunks (chunk_size= 1000, overlap = 20)**	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says COVID-19 triggered disruption to global supply chains at a scale not previously experienced, and it also led to remote working arrangements that increase reliance on IT systems and may raise fraud, conduct, operational, and other risks.** **Correct**	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: falling behind industry innovation, regulatory and legal uncertainty, cyberattacks and data leaks, bias and model errors, IP/copyright and malicious misuse risks, colleague misuse, customer discrimination or vulnerable customer misidentification, and added governance challenges from agentic AI systems.** **Correct**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to override, vary or impose contractual obligations between a UK bank, its holding company and group undertakings, and to amend the terms of contracts, including varying the maturity of debt instruments.** **Correct**	What is Lloyds' dividend policy for 2026? **-> I don't know.** **Correct**

Analysis: Both splitters performed almost identically because the document is mostly long continuous paragraphs. The RecursiveCharacterTextSplitter created slightly better chunk boundaries, but the overall difference was minimal.

- Retrieval

Chunk size, chunk overlap, top-k and temperature were selected because they are among the most influential parameters affecting retrieval quality and response generation in RAG systems.

Experiment 4 - top_k

- **Changed:** n_results = 1 / 3 / 7
- **Hypothesis:** The number of chunks retrieved would increase, but the relevancy would increase

Result table

top_k	Q1	Q2	Q3	Q4	Q5
1	[Q1_provision] What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £800 million** **Incorrect/Hallucinated**	**What does the document say about COVID-19?** **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	**What are the main risks Lloyds faces from artificial intelligence?** **-> Lloyds faces several AI-related risks: it can amplify existing risks and create new operational and competitive risks, require stronger privacy and ethical data processing controls, and create the risk of falling behind peers if others adopt AI faster or more effectively, leading to lost business opportunities.** Correct	**What powers does the Bank of England have under the Banking Act 2009?** **-> Under the Banking Act 2009, the Bank of England can amend contract terms, override events of default or termination rights, and override, vary, or impose contractual obligations between a UK bank, its holding company, and group undertakings for reasonable consideration.** Correct	**What is Lloyds' dividend policy for 2026?** **-> I don't know.** Correct
3	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** Correct	**What does the document say about COVID-19?** **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	**What are the main risks Lloyds faces from artificial intelligence?** **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection issues, ethical processing concerns, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer misidentification, governance challenges from agentic AI, and increased legal/regulatory risk as AI rules evolve. It also faces higher cyberattack and data leak risks.** Correct	**What powers does the Bank of England have under the Banking Act 2009?** **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private sector purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments and liabilities** **- placing a bank into temporary public ownership** **It also has powers to modify contractual arrangements, amend contract terms, override default or termination rights, and impose or vary obligations between a UK bank, its holding company, and group undertakings.** Correct	**What is Lloyds' dividend policy for 2026?** **-> I don't know.** Correct
7	**What was the total provision Lloyds recognised for the motor finance commission redress scheme?** **-> £1,950 million** Correct	**What does the document say about COVID-19?** **-> The document says COVID-19 triggered disruption to global supply chains at a scale not previously experienced.** Correct	**What are the main risks Lloyds faces from artificial intelligence?** **-> Lloyds faces several AI-related risks, including:** **- **Operational and competitive risk** from failing to keep pace with peers adopting AI faster** **- **Privacy, data protection, and ethical processing risks**** **- **Bias, errors, and discrimination**, including vulnerable customer misidentification** **- **Legal and regulatory risk** from rapidly changing AI rules across jurisdictions** **- **Cybersecurity and information leakage risks**, especially with third-party AI tools** **- **IP/copyright and malicious misuse risks**** **- **Governance challenges**, especially with more advanced "agentic" AI systems** Correct	**What powers does the Bank of England have under the Banking Act 2009?** **-> Under the Banking Act 2009, the Bank of England has powers to:** **- apply a mandatory write-down and conversion to Tier 1 and Tier 2 capital instruments** **- use a bail-in power over capital instruments and most unsecured liabilities** **- amend contractual terms, including varying debt maturity** **- override events of default or termination rights** **- override, vary, or impose contractual obligations between a UK bank, its holding company, and group undertakings** **These powers are used as part of the UK special resolution regime.** Correct	**What is Lloyds' dividend policy for 2026?** **-> I don't know.** Correct

Analysis: This had a noticeable impact on performance. Using top-k = 1 missed important context and returned £800 million instead of £1,950 million, while top-k = 3 consistently retrieved enough information without adding unnecessary context.

Experiment 5 - Semantic-only vs Hybrid Search

- **Changed:** pure semantic vector search vs. hybrid (semantic + keyword overlap)
- **Hypothesis:** Hybrid Search might pick up key details that semantic might miss.

Result table

Method	Q1	Q2	Q3	Q4	Q5
Semantic Only	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** **Correct**	**What are the main risks Lloyds faces from artificial intelligence?** **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection issues, ethical processing concerns, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer misidentification, governance challenges from agentic AI, evolving legal and regulatory uncertainty, and increased cyberattack and data leak risks.** **Correct**	**What powers does the Bank of England have under the Banking Act 2009?** **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private sector purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments and liabilities** **- placing a relevant entity into temporary public ownership** **It also has powers to modify or override contractual arrangements, including varying debt terms and overriding default or termination rights.** **Correct**	**What is Lloyds' dividend policy for 2026?** **-> I don't know.** **Correct**
Hybrid	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** **Correct**	**What are the main risks Lloyds faces from artificial intelligence?** **-> Lloyds faces several main AI risks: amplified risks across existing risk categories, new operational and competitive risks, privacy and data protection risks, ethical processing and governance risks, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer misidentification, evolving legal and regulatory risk, and increased cyberattack/data leak risk.** **Correct**	**What powers does the Bank of England have under the Banking Act 2009?** **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private purchaser or bridge bank** **- transferring assets to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- temporarily taking a bank into public ownership** **- modifying contractual arrangements, including overriding or varying contracts and certain default or termination rights** **Correct**	**What is Lloyds' dividend policy for 2026?** **-> I don't know.** **Correct**

Analysis: Hybrid search performed better because the document contains many specific legal terms and numerical facts. Combining semantic similarity with keyword matching improved retrieval for questions requiring exact information.

- Generation/LLM

Experiment 6 - Temperature

- **Changed:** temperature = 0 / 0.7 / 1.2
- **Hypothesis:** The answers will be more creative as temperature increases

Result table

Temperature	Q1	Q2	Q3	Q4	Q5
0	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** Correct	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection risks, ethical processing and governance challenges, potential discrimination and vulnerable customer misidentification, falling behind peers if it adopts AI too slowly, and increased cyberattack, data leak, legal, and regulatory risks from rapidly changing AI rules.** Correct	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- temporarily taking a bank into public ownership** **It also has powers to amend contract terms, override default or termination rights, and impose or vary contractual obligations to help a transferee or successor bank operate effectively.** Correct	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct
0.7	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** Correct	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data-protection risks, ethical and governance risks, discrimination and vulnerable customer misidentification, falling behind competitors if it adopts AI too slowly, evolving legal and regulatory uncertainty across jurisdictions, and increased cybersecurity/data-leak risks from AI-powered attacks.** Correct	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with failing banks through the special resolution regime, including:** **- transferring all or part of a bank's business or shares to a private purchaser** **- transferring business to a BoE-owned bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- imposing temporary public ownership** **It also has powers to modify contractual arrangements, override events of default or termination rights, and vary or impose obligations between a UK bank, its holding company, and group undertakings.** Correct	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct
1.2	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** Correct	What does the document say about COVID-19? **-> The document says that the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: increased operational and competitive risk, falling behind peers if it adopts AI too slowly, privacy and data protection risks, ethical processing and governance risks, discrimination and vulnerable-customer misidentification, and heightened legal/regulatory risk as AI rules evolve. It also faces greater**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to:** **- transfer all or part of a failing bank's business or shares to a private sector purchaser or bridge bank** **- transfer business to an asset management vehicle** **- bail in equity, capital instruments, and liabilities** **- modify or override contractual arrangements, including terms of debt instruments and default or termination rights**	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct

	Q3	Q4	Q5
?	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection risks, ethical processing and governance challenges, potential discrimination and vulnerable customer misidentification, falling behind peers if it adopts AI too slowly, and increased cyberattack, data leak, legal, and regulatory risks from rapidly changing AI rules.** **Correct**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- temporarily taking a bank into public ownership** **It also has powers to amend contract terms, override default or termination rights, and impose or vary contractual obligations to help a transferee or successor bank operate effectively.** **Correct**	What is Llo **-> I don't k** **Correct**
	cybersecurity and information-leak risks from AI-powered attacks. **Partial**	**- impose or vary contractual obligations between a UK bank, its holding company, and group undertakings** **I don't know if you want the full list beyond this excerpt.** **Correct**	

Analysis: Temperature had little effect on accuracy because the retrieved context already contained the answers. Temperature = 0 produced the most consistent and factual responses, making it the best choice for this document.

Experiment 7 - max_tokens

- **Changed:** max_tokens = 50 (forced truncation) / 500 (generous)
- **Hypothesis:** The output would be longer as max_tokens increases

Result table

max_tokens	Q1	Q2	Q3	Q4	Q5
50	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** **Correct**	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection risks, ethical processing and governance challenges, potential discrimination and vulnerable customer misidentification, falling behind peers if it adopts AI too slowly,** **Incorrect/Hallucinated**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including mandatory transfer of business or shares, transfer to a bridge bank, transfer to an asset management vehicle, bail-in** **Correct**	What is Lloyds' dividend policy for 2026? **-> I don't know.** **Correct**
500	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> £1,950 million** **Correct**	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** **Correct**	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing risks and new operational and competitive risks; privacy, ethical data processing, and compliance risks; falling behind peers if it adopts AI too slowly; discrimination and vulnerable customer misidentification; governance challenges from agentic AI; increased cyberattack and data leak risk; and evolving legal and regulatory risk across jurisdictions.** **Correct**	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private sector purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- placing the bank into temporary public ownership** **It also has powers to modify contractual arrangements, amend contract terms, override default or termination rights, and impose or vary obligations between a UK bank, its holding company, and group undertakings.** **Correct**	What is Lloyds' dividend policy for 2026? **-> I don't know.** **Correct**

Analysis: A limit of 50 tokens caused longer answers to be cut off. 500 tokens allowed complete responses while keeping factual accuracy the same.

Experiment 8 - Model Comparison

- **Changed:** gpt-5.4-mini vs gpt-4o
- **Hypothesis: The response from** gpt-5.4-mini would be more advanced and accurate due to being the newer model.

Result table

Model	Q	Q2	Q3	Q4	Q5
gpt-5.4-mini	**What was the total provision Lloyds recognised for the motor finance commission redress scheme?** **-> £1,950 million** Correct	What does the document say about COVID-19? **-> The document says the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection issues, ethical data processing concerns, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer misidentification, governance challenges from agentic AI, and increased legal/regulatory risk from rapidly changing AI rules. It also faces higher cyberattack and data leak risks.** Correct	What powers does the Bank of England have under the Banking Act 2009? **-> Under the Banking Act 2009, the Bank of England has powers to deal with and stabilise failing UK banks, including:** **- transferring all or part of a bank's business or shares to a private sector purchaser** **- transferring business to a bridge bank** **- transferring to an asset management vehicle** **- bailing in equity, capital instruments, and liabilities** **- placing the bank into temporary public ownership** **It also has powers to modify contractual arrangements, amend contract terms, override default or termination rights, and impose or vary obligations between a UK bank, its holding company, and group undertakings.** Correct	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct
gpt-4o	What was the total provision Lloyds recognised for the motor finance commission redress scheme? **-> The total provision Lloyds recognised for the motor finance commission redress scheme was £1,950 million.** Correct	What does the document say about COVID-19? **-> The document states that the COVID-19 pandemic triggered disruption to global supply chains at a scale not previously experienced.** Correct	What are the main risks Lloyds faces from artificial intelligence? **-> Lloyds faces several main AI risks: amplified existing operational and competitive risks, privacy and data protection issues, ethical data processing concerns, falling behind peers if it adopts AI too slowly, discrimination and vulnerable customer misidentification, governance challenges from agentic AI, and increased legal/regulatory risk from rapidly changing AI rules. It also faces higher cyberattack and data leak risks.**	What powers does the Bank of England have under the Banking Act 2009? **-> The Bank of England has the power to amend the terms of contracts, override events of default or termination rights, and override, vary, or impose contractual obligations between a UK bank, its holding company, and its group undertakings. Additionally, under the special resolution regime, the BoE can transfer all or part of a failing bank's business to a private sector purchaser, a bridge bank, or an asset management vehicle, conduct a bail-in, or place the bank into temporary public ownership.** Correct	What is Lloyds' dividend policy for 2026? **-> I don't know.** Correct

Analysis: Both models produced very similar answers. GPT-5.4-mini matched GPT-4o's accuracy while remaining consistent throughout the experiments, so it was chosen as the final model.

• Final Configuration Evaluation

Chunking

Rank	Setting	Why best for this document
1st	Chunk size = 1000, overlap = 20, RecursiveCharacterTextSplitter	Balanced context and precision. It consistently retrieved complete answers for long legal sections without adding unnecessary information.
2nd	Chunk size = 2000	Retrieved very detailed answers but sometimes included more

		information than needed because the document contains long legal paragraphs.
3rd	Chunk size = 500	Still found the correct answers, but smaller chunks occasionally split related legal information across multiple chunks.

Retrieval

Rank	Setting	Why best for this document
1st	Hybrid search + top_k = 3	Combined semantic meaning with keyword matching, which worked well for specific legal terms and numerical facts such as the motor finance provision and COVID-19 references.
2nd	Hybrid search + top_k = 7	Produced very complete answers but sometimes retrieved more context than necessary, increasing response length.
3rd	Semantic only / top_k = 1	Worked for broad questions but missed important context. With only one retrieved chunk it even returned **£800 million** instead of the correct **£1,950 million** because the document discusses multiple provisions.

Generation

Rank	Setting	Why best for this document
1st	Temperature = 0, max_tokens = 500, GPT-5.4-mini	Produced the most consistent and factual answers while avoiding hallucinations, which suits a regulatory document.
2nd	Temperature = 0.7	Responses remained accurate but wording became slightly more varied without improving correctness.
3rd	Temperature = 1.2	nswers were still mostly correct but became less concise and occasionally omitted details despite

		the higher creativity setting.

Overall Configuration

Rank	Setting	Why best for this document
1st	RecursiveCharacterTextSplitter (1000, overlap 20) + Hybrid Search (top_k = 3) + GPT-5.4-mini (temperature = 0, max_tokens = 500)	Produced the most consistent results across all five questions. It handled long legal paragraphs, cross-references and embedded numerical facts while correctly refusing to answer information that was not in the document.
2nd	RecursiveCharacterTextSplitter (2000, overlap 20) + Hybrid Search (top_k = 7) + GPT-5.4-mini (temperature = 0.7, max_tokens = 500)	Retrieved very detailed context, but responses were often longer than necessary because the larger chunks contained multiple related legal sections.
3rd	Fixed splitter (1000, overlap 20) + Semantic Search (top_k = 1) + GPT-5.4-mini (temperature = 1.2, max_tokens = 50)	Still answered several questions correctly, but retrieving only one chunk caused important context to be missed. This resulted in the incorrect **£800 million** answer for Q1 instead of the correct **£1,950 million**, and the lower token limit also truncated longer answers.

Recommended configuration for this document

Chunking

- RecursiveCharacterTextSplitter
- Chunk size = 1000
- Chunk overlap = 20

Retrieval

- Hybrid search
- top_k = 3

Generation

- GPT-5.4-mini
- Temperature = 0
- Max tokens = 500

Reflection

I expected larger chunks to perform best because they contain more context, but a chunk size of 1000 consistently gave the best balance between retrieving enough information and keeping answers focused. I also found that retrieval settings had a much bigger impact than generation settings. For example, retrieving only one chunk caused the system to return £800 million instead of the correct £1,950 million because the document contains multiple motor finance provisions. If I had more time, I would automate the evaluation using RAGAS and test the system on a different type of document to see whether these findings generalise.

.